Md Shahin Ali

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Biography

Md Shahin Ali is a PhD candidate and Graduate Research Assistant in the Roy J. Carver Department of Biomedical Engineering at the University of Iowa, where he earned his M.Sc. in Biomedical Engineering in May 2026; he received his B.Sc. (Engg.) in Biomedical Engineering from Islamic University, Kushtia, Bangladesh. Since August 2024, he has worked in the Laboratory of Quantitative and Dynamic MRI under Dr. Sajan Goud Lingala, developing Self-Supervised Spiral Variational Manifold Learning to image upper-airway collapse in obstructive sleep apnea. Previously, he was a research assistant in the Bio-Imaging Research Lab at Islamic University (2020–2024). His research interests include MR Image Reconstruction, Inverse Problems, Deep Learning, Machine Learning, Medical Image Processing, and Signal Processing. He also serves as a peer reviewer for several international journals.

Education

- **Doctor of Philosophy - The University of Iowa** — Iowa City, Iowa 52242, USA
  Biomedical Engineering (Ongoing) — *August 2024 - Present*

- **M.Sc. - The University of Iowa** — Iowa City, Iowa 52242, USA
  Biomedical Engineering — *August 2024 - May 2026*

- **B.Sc.(Engineering) - Islamic University, Bangladesh** — Kushtia 7003, Bangladesh
  Biomedical Engineering; GPA: 3.65 (Out of 4.00) → 4th among 43 peers — *January 2018 - February 2023*

- **Chuadanga Government College** — Chuadanga 7200, Khulna, Bangladesh.
  H.S.C.; Science; GPA: 4.33 (Out of 5.00) — *May 2014 - July 2016*

- **Damurhuda Pilot High School** — Damurhuda 7220, Chuadanga, Khulna, Bangladesh.
  S.S.C.; Science; GPA: 5.00 (Out of 5.00) — *January 2009 - March 2014*

Skills Summary

- **Languages**: Python, MATLAB, R, C, C++
- **Frameworks**: TensorFlow, Keras, scikit-learn, NumPy, Pandas, Matplotlib, Seaborn
- **Tools**: Microsoft Office Word, PowerPoint, Excel (Advanced User), Python Programming, LaTeX, SPSS
- **Soft Skills**: Leadership, Project Management, Event Management, Time Management, Writing, Public Speaking
- **Others**: Canva, Gmail, Skype, Social Media

Experience

- **Laboratory of Quantitative and Dynamic Magnetic Resonance Imaging** — The University of Iowa
  Graduate Research Assistant — *August 2024 - Present*
  - **Supervisor**: Sajan Goud Lingala, Associate Professor, Roy J. Carver Department of Biomedical Engineering.
  - **Research Area**: MR Image Reconstruction, Inverse Problems, Deep Learning, Signal Processing, & Image Processing.
  - **Funding**: Supported by NIH NHLBI grant R01 HL173483 and NIH-S10 instrumentation grant 1S10OD025025-01.

- **Graduate Peer-to-Peer (P2P) Mentor Program** — The University of Iowa
  Peer Mentor — *August 2026 - Present*
  - **Role**: Mentoring an incoming PhD student through their first year of graduate school.

- **Bio-Imaging Research Lab** — Islamic University, Bangladesh
  Research Assistant — *June 2020 - August 2024*
  - **Supervisor**: Md Khairul Islam, Associate Professor, Department of Biomedical Engineering.
  - **Research Area**: Medical Image Processing, Signal Processing, Machine Learning, Deep Learning, and Health Informatics.

- **IEEE Islamic University Student Branch** — Islamic University, Bangladesh
  Chair — *July 2021 - August 2022*
  - **Society**: IEEE Engineering in Medicine & Biology Society Islamic University Student Branch.
  - **Branch**: IEEE Islamic University Student Branch, Islamic University, Kushtia 7003, Bangladesh.

- **IEEE Islamic University Student Branch** — Islamic University, Bangladesh
  Program Executive — *December 2020 - December 2021*
  - **Branch**: IEEE Islamic University Student Branch, Islamic University, Kushtia 7003, Bangladesh.
- **Guidelines to be a Researcher to get a Scholarship** — Islamic University, Bangladesh
  Instructor and Trainer — *12 October 2022 - 15 November 2022*
  - **Course Outlines**: Check it here...
  - **Total Students Trained**: 50+
- **Swift Consultancy Ltd for Islamic University** — Dhaka 1207, Bangladesh
  Student Ambassador — *July 2021 - December 2021*
  - **Organization**: Swift Consultancy Ltd, Lalmatia, Dhaka 1207.

B.Sc. Project

- **A Hyper-tuned Vision Transformer Model with Explainable AI for Eye Disease Detection and Classification from medical images**: Developing a hyper-tuned Vision Transformer model with Explainable AI for Eye disease detection and classification from medical images can improve accuracy, speed up diagnosis, allocate resources more efficiently, and provide insights into the model's reasoning, ultimately improving patient outcomes in ophthalmology.
  Supervisor: Md Khairul Islam, Associate Professor, Department of Biomedical Engineering, Islamic University, Bangladesh.

Publications

Journal Articles

- **Md Shahin Ali**, Wahidul Alam, Mathews Jacob, Sarv Priya, Douglas Van Daele, Junjie Liu, and Sajan Goud Lingala, "Improved Upper-Airway Collapse Imaging in Obstructive Sleep Apnea Using Self-Supervised Spiral Variational Manifold Learning," ***Magnetic Resonance in Medicine (MRM)***, Oct. 2026. **Impact Factor: 3.2**. DOI: 10.1002/mrm.70644
- **Md Shahin Ali**, Md Khairul Islam, A Arjan Das, D S Duranta, Mst Farija Haque, Md Habibur Rahman, "A Novel Approach for Best Parameters Selection and Feature Engineering to Analyze and Detect Diabetes: Machine Learning Insights," ***BioMed Research International***, 2023. **Impact Factor: 2.9**. DOI: 10.1155/2023/8583210
- Md Manjurul Ahsan, **Md Shahin Ali**, Md Mehedi Hassan, Tareque Abu Abdullah, Kishor Datta Gupta, Ulas Bagci, Chetna Kaushal, Naglaa F Soliman, "Monkeypox Diagnosis with Interpretable Deep Learning," ***IEEE Access***, vol. 11, 2023. **Impact Factor: 4.2**. DOI: 10.1109/ACCESS.2023.3300793
- Md Manjurul Ahsan, **Md Shahin Ali**, Zahed Siddique, "Enhancing and Improving the Performance of Imbalanced Class Data Using Novel GBO and SSG: A Comparative Analysis," ***Neural Networks***, vol. 173, 2024. **Impact Factor: 6.0**. DOI: 10.1016/j.neunet.2024.106157
- Md Manjurul Ahsan, Muhammad Ramiz Uddin, **Md Shahin Ali**, Md Khairul Islam, Mithila Farjana, Ahmed Nazmus Sakib, Khondhaker Al Momin, Shahana Akter Luna, "Deep Transfer Learning Approaches for Monkeypox Disease Diagnosis," ***Expert Systems with Applications***, vol. 216, 119483, 2023. **Impact Factor: 9.4**. DOI: 10.1016/j.eswa.2022.119483
- S M Mahim, **Md Shahin Ali**, Md Olid Hasan, Abdullah Al Nomaan Nafi, Arefin Sadat, Shakib Al Hasan, Bryar Shareef, Md Manjurul Ahsan, Md Khairul Islam, "Unlocking the Potential of XAI for Improved Alzheimer's Disease Detection and Classification Using a ViT-GRU Model," ***IEEE Access***, vol. 12, 2024. **Impact Factor: 4.2**. DOI: 10.1109/ACCESS.2024.3351809
- Md. Maruf Hossain, **Md Shahin Ali***, Md. Mahfuz Ahmed, Md. Rakibul Hasan Rakib, Moutushi Akter Kona, Sadia Afrin, Md Khairul Islam, Md Manjurul Ahsan, Sheikh Md. Razibul Hasan Raj, Md Habibur Rahman, "Cardiovascular Disease Identification Using a Hybrid CNN-LSTM Model with Explainable AI," ***Informatics in Medicine Unlocked***, 2023. **CiteScore: 10.5**. DOI: 10.1016/j.imu.2023.101370
- **Md Shahin Ali**, Md Sipon Miah, Jahurul Haque, Md Mahbubur Rahman, Md Khairul Islam, "An Enhanced Technique of Skin Cancer Classification Using Deep Convolutional Neural Network with Transfer Learning Models," ***Machine Learning with Applications***, vol. 5, 100036, 2021. **Impact Factor: 6.1**. DOI: 10.1016/j.mlwa.2021.100036
- Md Khairul Islam, Md Mahbubur Rahman, **Md Shahin Ali**, S M Mahim, Md Sipon Miah, "Enhancing Lung Abnormalities Detection and Classification Using a Deep Convolutional Neural Network and GRU with Explainable AI: A Promising Approach for Accurate Diagnosis," ***Machine Learning with Applications***, vol. 13, 100492, 2023. **Impact Factor: 6.1**. DOI: 10.1016/j.mlwa.2023.100492
- Md Manjurul Ahsan, Tasfiq E. Alam, Mohd Ariful Haque, **Md Shahin Ali***, Rakib Hossain Rifat, Abdullah Al Nomaan Nafi, Md. Maruf Hossain, Md Khairul Islam, "Enhancing Monkeypox Diagnosis and Explanation through Modified Transfer Learning, Vision Transformers, and Federated Learning," ***Informatics in Medicine Unlocked***, vol. 45, 101449, 2024. **CiteScore: 10.5**. DOI: 10.1016/j.imu.2024.101449
- Md Khairul Islam, Md Mahbubur Rahman, **Md Shahin Ali**, S M Mahim, Md Sipon Miah, "Enhancing Lung Abnormalities Diagnosis Using Hybrid DCNN-ViT-GRU Model with Explainable AI: A Deep Learning Approach," ***Image and Vision Computing***, vol. 145, 104918, 2024. **Impact Factor: 5.0**. DOI: 10.1016/j.imavis.2024.104918

- Md Khairul Islam, Md Mahbubur Rahman, **Md Shahin Ali**, Md Sipon Miah, Md Habibur Rahman, "An Enhanced Technique of COVID-19 Detection and Classification Using Deep Convolutional Neural Network from Chest X-ray and CT Images," ***BioMed Research International***, vol. 2023, 6341259, 2023. **Impact Factor: 2.9**. DOI: 10.1155/2023/6341259

- Md Imran Hasan, **Md Shahin Ali**, Md Habibur Rahman, Md Khairul Islam, "Automated Detection and Characterization of Colon Cancer with Deep Convolutional Neural Networks," ***Journal of Healthcare Engineering***, vol. 2022, 5269913, 2022. **CiteScore: 2.9**. DOI: 10.1155/2022/5269913

- D U S Duranta, **Md Shahin Ali**, Abhilash Arjan Das, Md Mahbubur Rahman, Md Manjurul Ahsan, Md Sipon Miah, Md Khairul Islam, "Enhancing Atrial Fibrillation Detection Accuracy: A Wavelet Transform Filtered Single Lead ECG Signal Analysis with Artificial Neural Networks and Novel Feature Extraction," ***Machine Learning with Applications***, 2023. **Impact Factor: 6.1**. DOI: 10.1016/j.mlwa.2023.100472

- Mohammad Shafiul Alam, Muhammad Mahbubur Rashid, Ahmed Rimaz Faizabadi, Hasan Firdaus Mohd Zaki, Tasfiq E. Alam, **Md Shahin Ali**, Kishor Datta Gupta, Md Manjurul Ahsan, "Efficient Deep Learning-Based Data-Centric Approach for Autism Spectrum Disorder Diagnosis from Facial Images Using Explainable AI," ***Technologies***, vol. 11, 115, 2023. **Impact Factor: 5.2**. DOI: 10.3390/technologies11050115

Conference Proceedings

- **Md Shahin Ali**, Nicholas Thedens, Wahidul Alam, Mathews Jacob, Sarv Priya, Douglas Van Daele, Aygun Asgarli, Junjie Liu, and Sajan Goud Lingala, "Multi-Slice Dynamic Upper-Airway Collapse Imaging in Obstructive Sleep Apnea with a Spiral Unsupervised Generative Manifold Model," ***BMES 2026 Annual Meeting***, Orlando, Florida, USA, 21–24 October 2026. Poster # AA3.

- Junjie V. Liu, Aygun Asgarli, **Md Shahin Ali**, Nicholas E. Thedens, Douglas J. Van Daele, and Sajan G. Lingala, "0550 Dynamic Upper Airway MRI (DUAM) of OSA During Natural Sleep Is Suitable for Clinical Use," ***SLEEP***, Vol. 49, Issue Supplement_1, May 2026, Pages A244–A245. DOI: 10.1093/sleep/zsag091.0549

- Sajan G. Lingala, Nicholas E. Thedens, **Md Shahin Ali**, Douglas J. Van Daele, Aygun Asgarli, and Junjie V. Liu, "0551 Dynamic Upper Airway MRI (DUAM) of OSA May Be Better Than DISE for Detecting Complete Concentric Collapse," ***SLEEP***, Vol. 49, Issue Supplement_1, May 2026, Page A245. DOI: 10.1093/sleep/zsag091.0550

- **Md Shahin Ali**, Wahidul Alam, Bahaaaldin Mohammad, Mathews Jacob, Douglas Van Daele, Junjie Liu, and Sajan Goud Lingala. (2026). "Prospective validation of self-supervised spiral variational manifold learning for upper-airway collapse imaging." ***ISMRM Workshop on Data Sampling and Image Reconstruction***, Sedona, Arizona, USA, 11–14 January 2026. Abstract #00243.

- **Md Shahin Ali**, Wahidul Alam, Mathews Jacob, and Sajan Goud Lingala. (2025). "Sensitivity analysis of self-supervised variational manifold learning based accelerated dynamic upper-airway collapse MRI." ***ISMRM & ISMRT Annual Meeting & Exhibition***, Honolulu, Hawaii, USA, 10–15 May 2025. Abstract #2595.

- **Md Shahin Ali**, Md. Khairul Islam, Jahurul Haque, A Arjan Das, D S Duranta, Md Ariful Islam. (2021). "Alzheimer's Disease Detection Using m-Random Forest Algorithm with Optimum Features Extraction." ***1st International Conference on Artificial Intelligence and Data Analytics (CAIDA)***, pp. 1–6. IEEE Xplore, Riyadh, Saudi Arabia. DOI: 10.1109/CAIDA51941.2021.9425212

- Md. Khairul Islam, **Md Shahin Ali**, Md Mosahak Ali, Mst. Farija Haque, Abhilash Arjan Das, Md. Maruf Hossain, D S Duranta, Md Afifur Rahman. (2021). "Melanoma Skin Lesions Classification using Deep Convolutional Neural Network with Transfer Learning." ***1st International Conference on Artificial Intelligence and Data Analytics (CAIDA)***, pp. 48–53. IEEE Xplore, Riyadh, Saudi Arabia. DOI: 10.1109/CAIDA51941.2021.9425117

- **Md Shahin Ali**, Md. Maruf Hossain, Rubaya Nowrin, Md. Rakibul Islam, Md. Mahfuz Ahmed, Lamia Tasnim, Md. Mahbubur Rahman, and Md. Khairul Islam, "A Hyper-Tuned Vision Transformer Model with Explainable AI for Eye Disease Detection and Classification from Retinal Images," ***International Conference on Technology, Business, and Justice towards Smart Bangladesh (ICTBJ)***, Trishal, Mymensingh, Bangladesh, 5–6 June 2023. Oral Presentation.

Book Chapters

- Angona Biswas, Abdullah Nasim, **Md Shahin Ali**, Ismail Hossain, Azim Ullah, Sajedul Talukder, "Active Learning on Medical Image," in *Data Driven Approaches on Medical Imaging*, ***Springer***, pp. 51–67, 2023. DOI: 10.1007/978-3-031-47772-0_3

- Jannatul Nayem, Sayed Sahriar Hasan, Noshin Amina, Bristy Das, **Md Shahin Ali**, Md Manjurul Ahsan, Shivakumar Raman, "Few Shot Learning for Medical Imaging: A Comparative Analysis of Methodologies and Formal Mathematical Framework," in *Data Driven Approaches on Medical Imaging*, ***Springer***, pp. 69–90, 2023. DOI: 10.1007/978-3-031-47772-0_4

- Angona Biswas, Nasim Md Abdullah Al, **Md Shahin Ali**, Ismail Hossain, Azim Ullah, Sajedul Talukder, "Case Studies on X-ray Imaging, MRI and Nuclear Imaging," in *Data Driven Approaches on Medical Imaging*, ***Springer***, pp. 207–225, 2023. DOI: 10.1007/978-3-031-47772-0_10

Membership in Scientific Societies

- **International Society for Magnetic Resonance in Medicine (ISMRM)** – Trainee Member (2024 – Present)
- **Associated Professional Sleep Societies (APSS)** – Member (2026 – Present)
- **Biomedical Engineering Society (BMES)** – Member (2026 – Present)
- **Institute of Electrical and Electronics Engineers (IEEE)** – Student Member (2021 – 2024)

Reviewer

- Scientific Reports
- IEEE Access
- Artificial Intelligence Review (Springer)
- Biomedical Signal Processing and Control (Elsevier)
- Visual Computing for Industry, Biomedicine, and Art (Springer)
- Informatics in Medicine Unlocked (Elsevier)
- Machine Learning with Applications (Elsevier)
- International Journal of Imaging Systems and Technology (Wiley Online Library)
- Journal of Digital Imaging (Springer)
- The Imaging Science Journal (Taylor & Francis)
- MethodsX (Elsevier)
- MIET 2022: Series of Springer Nature

Honors and Awards

- **First Place, Graduate Student Poster Award** (BME), UI College of Engineering Research Open House, 2026.
- K. B. Chandran Graduate Student Travel Award, Roy J. Carver Dept. of BME, University of Iowa, 2025 and 2026.
- University Merit Scholarships (BSc), Islamic University, Kushtia-7003, Bangladesh.
- National Board Scholarship, Junior School Certificate Examination, 2011.
- National Board Scholarship, Primary School Certificate Examination, 2008.

References

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